

TrES-3b

R-0487

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_230925 · created 2026-07-20T08:36:23+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460212.6515 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.014
Transit depth (Rp/Rs)^2	3.85 +/- 0.55 [%]
Orbital Inclination (inc)	82.46 +/- 1.01
Airmass coefficient 1 (a1)	1.052 +/- 0.013
Airmass coefficient 2 (a2)	-0.043 +/- 0.022
Scatter in the residuals of the lightcurve fit is	1.22 %
Best Comparison Star	None
Optimal Aperture	4.24
Optimal Annulus	6.71
Transit Duration (day)	0.0626 +/- 0.0082

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.196 $\pm$ 0.014 vs 0.16309 (deviation 2.35 σ)
Duration fit vs published	0.0626 d vs 0.0567 d (deviation 0.72 σ)
Mid-time O-C	-5.4 min
Depth SNR	14.0
Residual scatter	1.22 %

Light curve

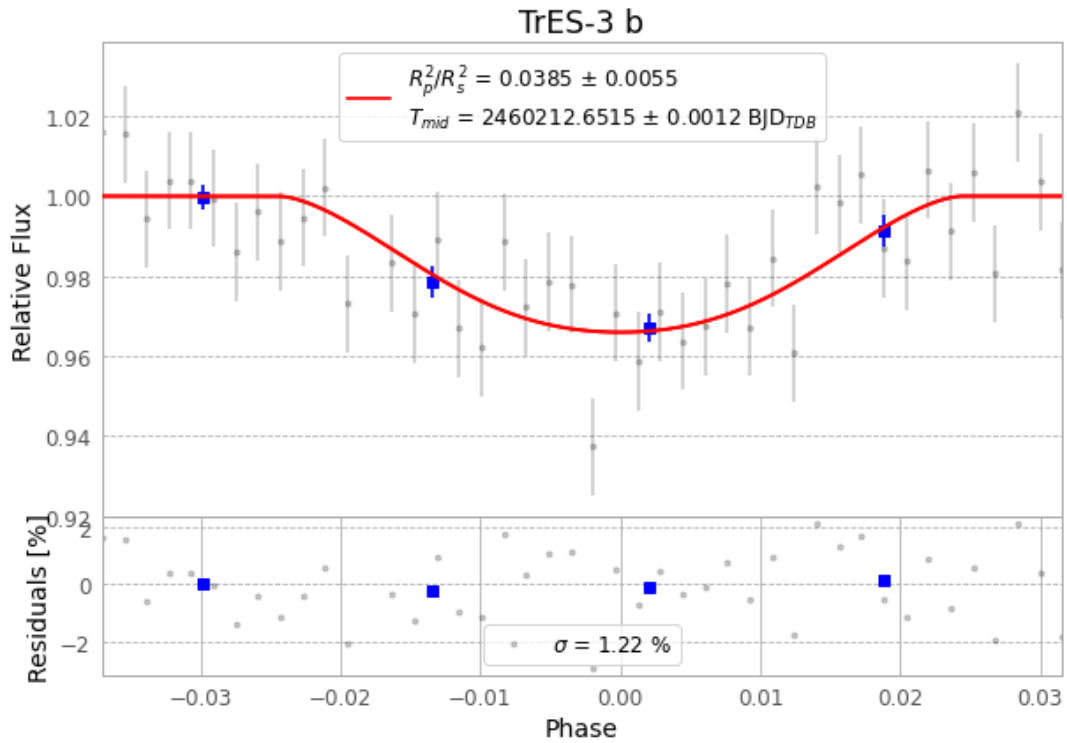


Figure 1 — FinalLightCurve_TrES-3 b_2023-09-24.png (SHA-256 8d10a14b046df189...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [270, 181], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 88b7dcee5be61565...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

44 FITS frames (29 MB), TRES-3230925022715.FITS → TRES-3230925043615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-230925.log (6a2c33e9c3ec...), FinalLightCurve_TrES-3 b_2023-09-24.png (8d10a14b046d...), FinalLightCurve_TrES-3 b_2023-09-24.csv (fe0d942e4d66...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 08:36Z.